

Technology Stack

Date	29-06-2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML & CSS	Provides a responsive design and component reusability for a smooth user experience

architecture and provide a brief justification for your choice.

Technology Stack Details :

The development of the "AI-based Credit Card Approval System" utilizes a robust and scalable technology stack designed to ensure efficient data processing, accurate credit risk prediction, and a seamless user experience.

Frontend Development: The user interface is built using React.js, which ensures a responsive and interactive experience for applicants, allowing them to track their application status in real-time.

Backend Architecture: Python (with the Flask/Django framework) is employed for the backend, as it provides excellent integration capabilities with AI/ML libraries for processing complex credit scoring algorithms.

Data Management: PostgreSQL is utilized as the primary database to ensure structured, reliable, and secure storage of applicant financial profiles and credit history.

AI/ML Logic: The core credit assessment is powered by Scikit-learn, which executes the machine learning models required to predict creditworthiness based on historical data.

Deployment and DevOps: The application is hosted on AWS (Amazon Web Services) for its high availability and scalability, with GitHub and Jenkins used for version control and CI/CD pipelines to ensure automated testing and deployment.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask or Django)	Excellent support for AI/ML libraries and fast processing of data.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	PostgreSQL	Strong relational mapping to handle structured applicant and credit data reliably.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	AWS (Amazon Web Services)	Offers easy scalability and cost-effective hosting options for the web application.

5	Version Control & CI/CD	GitHub & Jenkins	Enables seamless team collaboration and automated testing/deployment.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Scikit-learn (for AI model)	Used for building and executing the credit scoring prediction logic.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		